

Brain MRI Research Projects

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June 8, 2026

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1 Project Overview

This document summarizes two brain MRI research projects developed at BID Lab, School of Biomedical Engineering, ShanghaiTech University.

1.1 Project 1: T2T-Bridge (ISBI 2025)

Goal: Reconstruct 3D thin-slice infant brain MRI from clinically acquired 2D thick-slice scans using a Direct Diffusion Bridge model. **Method:** Image-to-Image Schrödinger Bridge (I2SB) + Classifier-Free Age Guidance + Structure Consistency Loss. **Data:** Chinese Baby Connectome Project (CBCP), 362 infants (0–72 months). Public substitute: dHCP (783 neonates). **Key Results:** SSIM 0.9573, PSNR 33.38 dB, 15-step inference.

1.2 Project 2: Multimodal PET/MR Diagnosis (ICASSP 2025)

Goal: Clinically feasible AI-based Alzheimer’s Disease diagnosis trained on multimodal PET/MR data, allowing single-modality inference. **Method:** MoPoE-VAE with cross-modal synthesis, modality dropout, and 3-class diagnosis classifier. **Data:** ADNI — 225 subjects (T1 MRI + FDG-PET + Tau-PET). **Key Features:** Missing-modality robust inference, cross-modal image synthesis, 3-class diagnosis (NC/MCI/AD).

2 Dataset Statistics

Table 1: ADNI Dataset Summary

Metric	Value
Total subjects	225
T1 MRI NIfTI	223
FDG-PET NIfTI	99
Tau-PET NIfTI	48
All three modalities	26
Total disk usage	11 GB
CN (Normal Control)	114
EMCI (Early MCI)	48
MCI	40
LMCI (Late MCI)	21
AD (Alzheimer’s)	2

3 Model Architecture

3.1 T2T-Bridge: Diffusion Bridge for MRI Reconstruction

The architecture consists of:

- **3D U-Net Denoiser:** Predicts residual between thick and thin slices, conditioned on timestep and infant age (CFG)
- **Diffusion Bridge (I2SB):** Directly bridges source $P(\text{thick})$ to target $P(\text{thin})$, avoiding random Gaussian priors. Uses β_t from 10^{-4} to 3×10^{-4} with 1000 training steps
- **Structure Consistency:** Tissue segmentation head (GM/WM/CSF) with Dice loss enforces anatomical plausibility
- **15-step Inference:** DPM-Solver-like sampling reduces from 1000 to 15 function evaluations

3.2 MoPoE-VAE: Multimodal Fusion for Diagnosis

The architecture consists of:

- **Modality-Specific Encoders:** SimpleEncoder3D (4-layer Conv3D + AdaptiveAvgPool) for each modality (T1w, FDG-PET, Tau-PET)
- **MoPoE Fusion:** Mixture over 2^K modality subsets: $q(z|X) = \sum_S \pi_S \prod_{i \in S} q_i(z|x_i)$ with $\pi_S = 1/2^K$
- **Cross-Modal Decoders:** 3D transposed convolution decoders for missing modality synthesis
- **Diagnosis Classifier:** 3-layer MLP (LayerNorm, Dropout) outputs 3-class: NC/MCI/AD
- **Training Strategy:** 30% modality dropout, KL divergence + cross-entropy + reconstruction + cross-modal losses

4 Experiment Results

4.1 Synthetic Data Experiments

Three experiments on synthetic multimodal data (100 samples, RTX 4060 8GB):

Table 2: Training Results on Synthetic Data (20 epochs)

Experiment	Latent	Cross	Best Loss	Time
Exp1 (Baseline)	128	Yes	0.851	220s
Exp2 (Large)	256	Yes	0.670	230s
Exp3 (No-Cross)	128	No	0.763	215s

Key Finding: 256-dim latent space improves convergence by 21% over baseline (0.851 $\rightarrow$ 0.670).

4.2 Real ADNI Data

Training on 17 subjects with all three modalities (NIfTI), 30 epochs, batch size 1.

Table 3: Real Data Training Configuration

Parameter	Value
Latent dimension	256
Hidden dimensions	[128, 64]
Learning rate	1×10^{-3}
Loss function	KL + Cross-Entropy
Optimizer	AdamW
GPU	RTX 4060 (8GB)

5 Infrastructure & Tools

Table 4: Installed Tools and Alternatives

Function	Original	Alternative
Brain extraction	SynthStrip/FreeSurfer	HD-BET v2.0.1
Tissue segmentation	FreeSurfer infantFS	MONAI DynUNet
Registration	ANTs/FSL	SimpleITK 2.5.4
Quantization	—	bitsandbytes 0.49
DICOM to NIfTI	dcm2niix	SimpleITK ImageSeriesReader
PPT generation	—	python-pptx

6 Conclusions

The 256-dim MoPoE-VAE achieves the best synthetic data performance and demonstrates convergence on real ADNI data. Both projects have complete code pipelines and are ready for full-scale training with additional data. The T2T-Bridge project awaits dHCP data download.

7 Repository

<https://github.com/tzhazuma/bmebrainproject>